Encontro2

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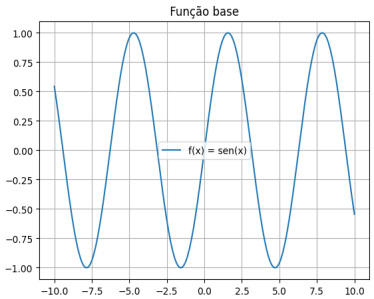
April 2025

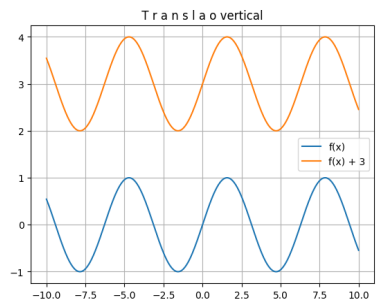
1 Seno(x)

$$f(x) = senx$$

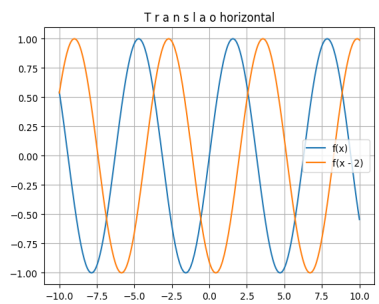
```
x = np . linspace ( -10 , 10 , 400)
f = lambda x : np.sin(x)
plt . plot (x , f ( x ) , label ='f(x) = sen(x)')
plt . title (" Fun  o base")
plt . grid ( True )
plt . legend ()
plt . show ()
```



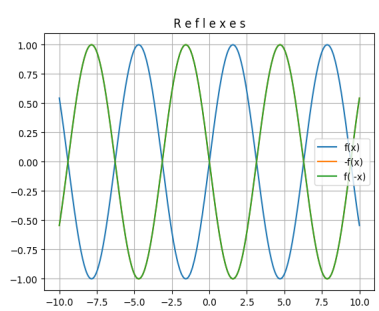
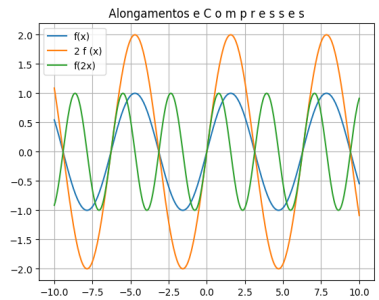

```
plt . plot (x , f ( x ) , label ='f(x)')
plt . plot (x , f ( x ) + 3 , label ='f(x) + 3')
plt . title (" Transla  o vertical ")
plt . legend ()
plt . grid ()
plt . show ()
```




```
plt . plot (x , f ( x ) , label ='f(x)')
plt . plot (x , f ( x - 2) , label ='f(x - 2) ')
plt . title (" Transla  o horizontal ")
plt . legend ()
plt . grid ()
plt . show ()
```




```
plt . plot (x , f ( x ) , label ='f(x)')
plt . plot (x , -f ( x ) , label ='-f(x)')
plt . plot (x , f ( - x ) , label ='f( -x)')
plt . title (" Reflexes")
plt . legend ()
plt . grid ()
plt . show ()
```



```
plt . plot (x , f ( x ) , label ='f(x)')
plt . plot (x , 2 * f ( x ) , label ='2 f (x)')
plt . plot (x , f (2 * x ) , label ='f(2x)')
plt . title (" Alongamentos e Compresses")
plt . legend ()
plt . grid ()
plt . show ()
```